



Diabetes Prediction using Machine Learning

A machine learning project that predicts whether a person is likely to have diabetes based on health-related parameters. This project was created to understand the complete machine learning workflow, including data preprocessing, model training, evaluation, and prediction using Python.



Project Overview

Diabetes is one of the most common chronic diseases worldwide. Early prediction can help in timely diagnosis and better healthcare decisions. This project uses the **Pima Indians Diabetes Dataset** to build a classification model capable of predicting whether a person is diabetic based on medical information.

The project demonstrates each step involved in building a machine learning model, making it suitable for beginners who want to learn practical implementation using Scikit-learn.



Project Structure

```
Diabetes-Prediction-using-ML/  
|  
├── Diabetes_Prediction_using_ML.ipynb  
├── diabetes.csv  
└── README.md
```



Dataset

The dataset contains medical information of female patients and is commonly used for binary classification tasks.

Features

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI (Body Mass Index)
- Diabetes Pedigree Function
- Age

Target Variable

- **0** → Non-Diabetic
 - **1** → Diabetic
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Technologies Used

- Python
 - Jupyter Notebook
 - NumPy
 - Pandas
 - Matplotlib
 - Scikit-learn
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Machine Learning Model

This project uses the **Support Vector Machine (SVM)** algorithm with a linear kernel for binary classification.

The workflow includes:

- Importing libraries
 - Loading the dataset
 - Data exploration
 - Feature scaling using StandardScaler
 - Splitting the dataset into training and testing sets
 - Training the SVM classifier
 - Evaluating model accuracy
 - Predicting diabetes for new patient data
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Getting Started

Clone the Repository

```
git clone https://github.com/your-username/Diabetes-Prediction-using-ML.git
```

Install Required Libraries

```
pip install numpy pandas matplotlib scikit-learn jupyter
```

Run the Notebook

```
jupyter notebook Diabetes_Prediction_using_ML.ipynb
```

Run all the notebook cells in order to reproduce the results.



Future Enhancements

Some improvements that can be added to this project include:

- Comparing multiple machine learning algorithms
 - Hyperparameter tuning using GridSearchCV
 - Confusion Matrix
 - Precision, Recall and F1-Score
 - ROC Curve
 - Cross-validation
 - Deploying the model using Streamlit or Flask
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Learning Outcomes

Working on this project helped me gain hands-on experience in:

- Data preprocessing
 - Feature scaling
 - Binary classification
 - Machine learning model training
 - Model evaluation
 - Predictive analytics using Scikit-learn
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Author

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Passionate about building practical projects and continuously improving my skills in data-driven technologies.

★ Thank you for visiting this repository. Feedback and suggestions are always welcome!